

RareArena: a comprehensive benchmark dataset unveiling the potential of large language models in rare disease diagnosis

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Rare diseases pose a substantial clinical and public health burden, with timely and accurate diagnoses remaining a formidable challenge in many countries and settings. Large language models (LLMs) have the potential to enhance the screening and diagnosis of rare diseases. However, existing evaluations of such capacities of LLMs are insufficient due to the low sample sizes, poor coverage of different rare diseases, and weak real-world clinical relevance. To address these issues, we developed an automated data collection pipeline, RareArena, to curate a comprehensive rare disease dataset from PubMed Central case reports. Two tasks were considered: rare disease screening (RDS), which involved rare disease recognition before confirmatory diagnostic tests; and rare disease confirmation (RDC), which required diagnosis with complete information. The constructed RDS task comprised 49 760 cases of 4597 rare diseases, whereas the RDC task comprised 22 901 cases of 3522 rare diseases. Human evaluations by two physicians across three dimensions—leakage, fidelity, and complexity—validated the high quality of our dataset. We systematically benchmarked ten state-of-the-art LLMs of varying sizes (GPT-4o, Llama3·1-70B, Llama3·1-8B, Qwen2·5-72B, Qwen2·5-7B, Gemma2-9B, Phi3-7B, Llama3-70B-UltraMedical, Llama3-8B-UltraMedical, and MedLlama-7B) and found that GPT-4o achieved the best performance in both the tasks (RDS: 33·1% top-1 recall, 56·9% top-5 recall; RDC: 64·2% top-1 recall, 85·9% top-5 recall). Additionally, GPT-4o performed better on genetically inherited diseases than on non-genetically inherited diseases in both the tasks, with its highest performance observed in systemic or rheumatologic diseases (RDS: 48·5% top-1 recall, 76·5% top-5 recall; RDC: 76·0% top-1 recall, 93·5% top-5 recall). To our knowledge, our dataset represents the largest rare disease diagnostic benchmark to date, featuring broad and deep disease coverage and reflecting clinically relevant scenarios. RareArena highlights the potential of LLMs in rare disease screening and diagnosis, providing a foundation for improved care for individuals with rare diseases around the world.

Introduction

Rare diseases pose a substantial clinical and public health burden. Although individually rare, these diseases are collectively common, affecting approximately 6% of the global population and imposing societal costs nearly ten times higher than those of common diseases.^{1–5} Diagnosing rare diseases can be challenging due to the individuals' complex symptoms and physicians' insufficient knowledge of rare diseases.^{6,7} Consequently, correct diagnosis of rare diseases is often substantially delayed, with nearly 20% of individuals diagnosed 10 years after the initial onset of symptoms.^{8,9}

Of late, large language models (LLMs) have shown immense potential across multiple clinical practices, including patient triage and stratification,^{10–12} pharmacy medication guidance,¹³ and diagnosis support.^{14,15} The capacity of LLMs' to encode vast medical knowledge positions them as promising tools to assist physicians in both early screening and diagnosis of rare diseases. However, such potential of LLMs remains underexplored, primarily due to the paucity of comprehensive and clinically relevant benchmark datasets.

Previous evaluations of rare disease diagnostic capacities of LLMs have been constrained by small case numbers and, more importantly, by low disease coverage: ranging from a few to a maximum of 421 diseases.^{16–20} However, the full rare disease spectrum encompasses 5000 to 10 000 diseases.⁵ Consequently, these findings are not

generalisable and do not characterise the overall capabilities of LLMs in rare disease diagnosis. Another key limitation of the existing datasets is their inadequate reflection of real-world clinical scenarios.^{19–21} First, several studies have pointed out that existing medical benchmarks, mostly derived from medical examination questions, do not align well with clinical practice.^{22–24} For example ReDis-QA,¹⁶ a rare disease benchmark consisting of 1360 multiple-choice questions covering symptoms, causes, and diagnoses, still left a substantial gap between examination questions and real-world applications. Second, multiple studies focusing on benchmarking rare disease diagnosis used clinical phenotypes rather than free-text patient notes as the model input.^{18–20} This approach diverges from real-world scenarios, wherein most cases are documented in free text. Third, the complexity of real-world diagnostic procedures is not faithfully reflected in the existing benchmarks. Details such as negative symptoms that aid differential diagnosis or irrelevant positive symptoms that require careful distinction could be neglected in examination questions or phenotypes.

To address these limitations, we aimed to develop a comprehensive rare disease diagnostic benchmark featuring a larger case scale, broader disease coverage, and better alignment with clinical scenarios than the existing ones. We built an automated pipeline to extract numerous case reports from PubMed Central, screen for those focused on rare disease diagnosis, and transform them into a qualified

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diagnostic benchmark. For a more thorough analysis, we formulated two tasks with different levels of information abundance (figure 1): one without diagnostic test results and the other with the results included. These tasks reflect two potential clinical scenarios in the rare disease diagnostic procedure: early suspicion and diagnosis confirmation. We further benchmarked the most advanced LLMs on these two tasks and conducted systematic subgroup analyses to elucidate the variation in the diagnostic performance of LLMs across different rare disease categories.

Methods

Dataset construction

The dataset construction involved several steps: case extraction, case filtering, mapping rare diseases, truncation and rephrasing, and leakage detection. The procedure was primarily implemented using GPT-4o (2024-05-13 version) and is summarised in figure 1. The entire pipeline, including all the prompts for GPT-4o, was carefully designed and refined with medical expertise (by HC and SH).

We first extracted patient summaries from PubMed Central by reproducing PubMed Central-Patients²⁵ with the latest PubMed Central release (June, 2024; 5.8 million articles). This process primarily involved identifying and extracting paragraphs with potential case reports and filtering them using regular expressions. 250 294 cases were extracted as our case candidate pool.

The cases were manually reviewed (by HC, SH, and XJ) and comprehensive rules were designed to identify cases primarily focusing on rare disease diagnosis, as detailed in the appendix (p 19). After filtering, 104 000 cases remained. Additionally, the final diagnosis of the cases was extracted to serve as ground-truth labels for our benchmark. The case filtering and diagnosis extraction were implemented using GPT-4o, with corresponding prompts detailed in the appendix (p 20).

The definition of rare diseases varies across different geographical regions.⁵ To ensure consistency, we adopted Orphanet, a well acknowledged rare disease knowledge graph, as our reference. This open-source database includes more than 6000 rare diseases and provides comprehensive information on their subfield classification, clinical phenotypes, inheritance patterns, and epidemiology.^{26,27} We defined rare diseases as those listed in Orphanet. Therefore, each case's diagnosis was mapped to Orphanet and cases without adequate mapping were excluded. To tackle medical term diversity and achieve accurate mapping, embeddings for each diagnosis were generated using CODER,²⁸ a state-of-the-art bidirectional encoder representations from transformers (BERT)-based medical term encoder. The top 10 most similar disease terms were then retrieved from Orphanet on the basis of cosine similarities of the embeddings.

Subsequently, GPT-4o was used to identify the correct mapping among the top 10 diseases. Of the 104 000 cases after filtering, 57 000 diagnoses were successfully mapped to Orphanet diseases. We manually inspected 100 mapped diagnoses and observed an accuracy of 94%. Further details about term embeddings, retrieval, and the mapping process can be found in the appendix (p 1), whereas the prompts are available in the appendix (p 21). We did not use the mapped Orphanet diseases as ground-truth labels, as the diagnoses extracted by GPT-4o from the original cases were expected to be more specific and accurate.

The original cases could not be taken directly as model input since the texts already contained the diagnosis and potentially some highly indicative evidence. We first prompted GPT-4o to extract the tests for the diagnosis. Diagnostic tests were usually defined as a clinical approach that provides a high level of confidence in identifying an individual's diagnosis.^{29,30} In our study, diagnostic tests were defined as those that directly confirm or strongly suggest the diagnosis, such as whole-genome sequencing for genetic diseases, histopathological analysis for rare tumours, and pathogen detection for rare infections. The definition was also specified in our prompt (appendix p 22). Based on whether to include the diagnostic test results in the model inputs, we formulated two task settings and applied different truncation strategies to assess the model's capabilities more comprehensively. In the first setting, rare disease screening (RDS), the cases were truncated up to the point just before any diagnostic tests were performed. In the second setting, rare disease confirmation (RDC), the cases were truncated up to the point just before the final diagnosis. Cases without specific diagnostic tests were only included in the RDS task. GPT-4o was further prompted to rephrase the cases to include only the factual clinical findings, removing all subjective analyses, including evaluations of the patient's condition, rationale for clinical decisions, and suspected diagnoses (shown in appendix p 2). The rephrasing was intended to minimise potential diagnosis leakage. In addition, the rephrasing also served the purpose of avoiding data contamination, considering that most LLMs have encountered PubMed Central articles during pretraining. The corresponding prompts are listed in the appendix (p 22).

If the transformed cases still presented diagnosis leakage, then we conducted leakage detection using GPT-4o and removed any cases with potential leakage. The prompts used for leakage detection can be found in the appendix (p 23).

Human evaluation of dataset quality

Ten cases were randomly selected and jointly reviewed by two physicians (HC and SH) to develop a consistent ruleset for case quality evaluation (appendix p 3). HC and SH then independently scored a random sample of 100 cases

See Online for appendix

For more on Orphanet see
<https://www.orpha.net/>

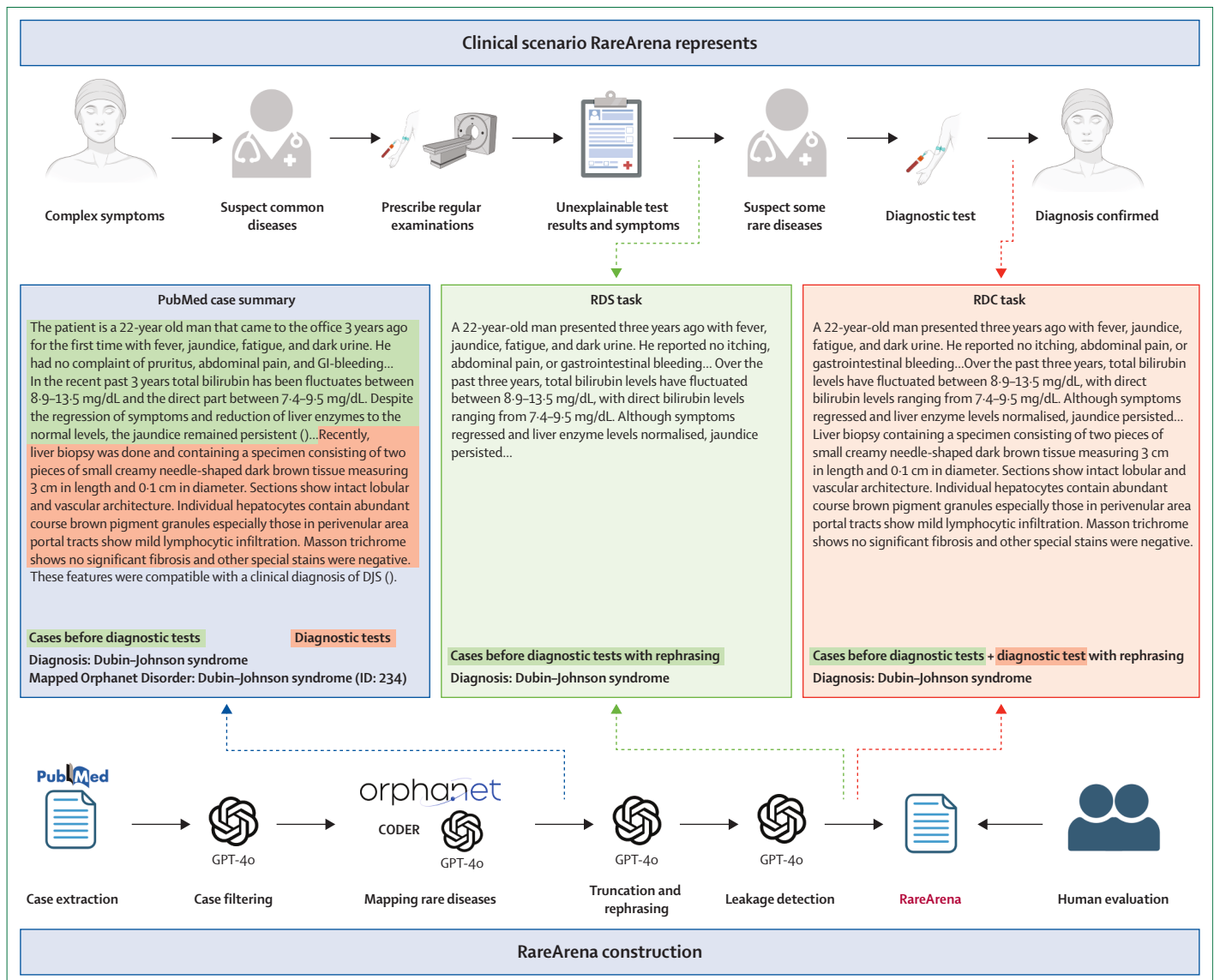


Figure 1: RareArena construction workflow and the clinical scenario it represents

Illustration of a typical diagnostic journey for patients with rare diseases and how RareArena operationalises this process into two evaluation tasks: RDS and RDC. RDS targets the pre-test stage, where symptoms and routine examinations are available but definitive diagnostic testing has not yet been performed. RDC targets the post-test stage, where diagnostic test results are available. RareArena is curated from PubMed case reports. Each original report is split into two segments: the clinical narrative before diagnostic testing (highlighted in green) and the diagnostic test findings (highlighted in orange). Construction pipeline for RareArena is also summarised, from case extraction to human evaluation. DJS=Dubin-Johnson syndrome. RDC=rare disease confirmation. RDS=rare disease screening.

(50 each for RDS and RDC). Any case with a disagreement was assessed by a third reviewer (XJ). The cases were evaluated from three distinct dimensions. The first dimension assessed whether the modified cases leak the diagnosis. The second dimension evaluated the fidelity of the case-generation process. Since the process was conducted automatically by GPT-4o, the accuracy of diagnosis extraction, diagnostic test identification, and case rephrasing was evaluated. The third dimension evaluated complexity with five subdimensions. The first two aimed to assess whether the case preserved real-world complexity, given that case reports are often rewritten to emphasise evidence most

relevant to the diagnosis. Specifically, we assessed whether the cases included pertinent negatives and confounding positives, which are commonly encountered in real clinical settings (as exemplified in appendix p 24). The remaining three were more objective dimensions: multimorbidity, polypharmacy, and extra health-care encounters (details in appendix p 3 and pp 25–26).

Benchmarking model performance

The models were provided with the transformed cases as input and outputted the top five most likely diagnoses. Due

to high computational costs, we primarily adopted a zero-shot setting and reported results using chain-of-thought and few-shot prompting on a randomly sampled subset of 500 cases (all evaluated at a temperature of 0.1). For few-shot prompting, we used three fixed, randomly sampled demonstrations (distinct from the 500 evaluation cases). The same prompts were used for all models (appendix p 27). Top-1 recall and top-5 recall were selected as the official metrics.

Aligning the model outputs with the ground-truth diagnoses was not straightforward due to the potential presence of synonyms and hypernyms. Therefore, we used GPT-4o to identify whether the true diagnosis appeared in the top 1 or top 5 model outputs, similar to approaches in previous studies.^{18,19} We developed a 3-graded scoring system: a score of 2 for synonym matches, 1 for hypernym matches, and 0 otherwise. GPT-4o achieved satisfactory scoring agreement with two human reviewers (SH and HC) on 100 randomly sampled cases. GPT-4o also showed higher consistency than another cutting-edge LLM, DeepSeek V3, indicating satisfactory reliability. The prompt used for scoring is provided in the appendix (p 28). The cases used to evaluate GPT-4o as a judge, assess zero-shot and few-shot performance, conduct quality evaluation, and verify Orphanet mapping accuracy were independently and randomly sampled.

For baseline models, we included four groups of state-of-the-art LLMs: a leading proprietary model, wherein proprietary refers to no publicly available weights: GPT-4o; 70B general-domain open-source models, wherein open-source refers to accessible weights: Llama3.1-70B and Qwen2.5-72B;³¹ 7B general-domain open-source models: Llama3.1-8B, Qwen2.5-7B,³¹ Phi3-7B,³² and Gemma2-9B;³³ and biomedical LLMs: Llama3-70B-UltraMedical and its 8B alternative³⁴ and MedLlama-7B. Although we attempted to include MedFound-176B, it was excluded due to its inability to reliably follow our task instructions. These models not only show cutting-edge performance among current LLMs, but are also widely used in clinical practice, including in diagnostic applications.^{16,35–37} Additionally, GPT-o1, which features an internalised chain-of-thought and cutting-edge reasoning capacities, was evaluated on a subset of 500 cases in comparison to GPT-4o, due to its high application programming interface (API) costs. The evaluations of GPT-4o, Llama3.1-70B/8B, Qwen2.5-72B/7B, Gemma2-9B, and Phi3-7B were conducted in August, 2024, whereas Llama3-70B-UltraMedical and its 8B alternative were evaluated in April, 2025, and MedLlama-7B in September, 2025.

Besides benchmarking the model's performance, we also conducted several secondary analyses. First, we compared performance on cases published before versus after each model's knowledge cut-off date to evaluate whether the model's performance could be influenced by potential training-data leakage. Second, we performed subgroup analyses to characterise performance heterogeneity across disease categories. We examined three key disease attributes including clinical subfield, genetic background, and

prevalence that were assigned at the case level based on the mapped Orphanet disease concept. These attributes were derived from the Orphanet database, where disease characteristics are systematically recorded. The statistical methods used to analyse the model's performance and the corresponding secondary analyses are described in the appendix (p 1).

Role of the funding source

The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication. TYW and SY were responsible for the decision to submit the manuscript for publication.

Results

RareArena dataset

RareArena comprised 49 760 cases for the RDS task and 22 901 cases for the RDC task, notably surpassing existing benchmarks (appendix p 4). The pronounced difference in case numbers between the two tasks was primarily attributed to the diagnosis leakage in the diagnostic test results of some cases, which were excluded from the RDS. Cases in RDS had an average word count of 203.7, whereas those in RDC averaged 261.7 (appendix p 13). The sex distribution in RareArena cases, as reported by the original case reports, was nearly equal (RDS: 24 432 [49.1%] of 49 760 female and 25328 [50.9%] of 49 760 male individuals; RDC: 11 611 [50.7%] of 22 901 female and 11 290 [49.3%] of 22 901 male individuals). More than 75% of RareArena cases involved adults (>18 years; 39 408 [79.2%] of 49 760 for RDS and 17 737 [77.5%] of 22 901 for RDC), whereas less than 25% involved children or newborns (10 352 [20.8%] of 49 760 for RDS and 5164 [22.5%] of 22 901 for RDC; figure 2). More than 80% of RareArena cases were non-genetically inherited diseases or diseases with unclear genetic contributions (41 549 [83.5%] of 49 760 for RDS and 18 442 [80.5%] of 22 901 for RDC; figure 2). RareArena encompassed cases across diverse medical subfields. The top three subfields by case prevalence in RDS were neoplastic diseases (10 764 [21.6%] of 49 760), infectious diseases (7109 [14.3%] of 49 760), and systemic or rheumatologic diseases (6040 [12.1%] of 49 760). In the RDC task, the leading subfields were neoplastic diseases (5060 [22.1%] of 22 901), systemic or rheumatologic diseases (3336 [14.6%] of 22 901), and neurological diseases (2791 [12.2%] of 22 901). Conversely, fields with the fewest cases included otorhinolaryngologic diseases (274 [0.6%] of 49 760), odontologic diseases (241 [0.5%] of 49 760), and disorders due to toxic effects (118 [0.2%] of 49 760) in RDS; and otorhinolaryngologic disorders (137 [0.6%] of 22 901), toxic effect disorders (38 [0.2%] of 22 901), and gynaecological or obstetric diseases (125 [0.5%] of 22 901) in RDC.

RareArena encompassed 4597 disorder concepts for the RDS tasks and 3522 disorder concepts for the RDC tasks, with a median case count per disorder of 3 (Q1: 1; Q3: 3;

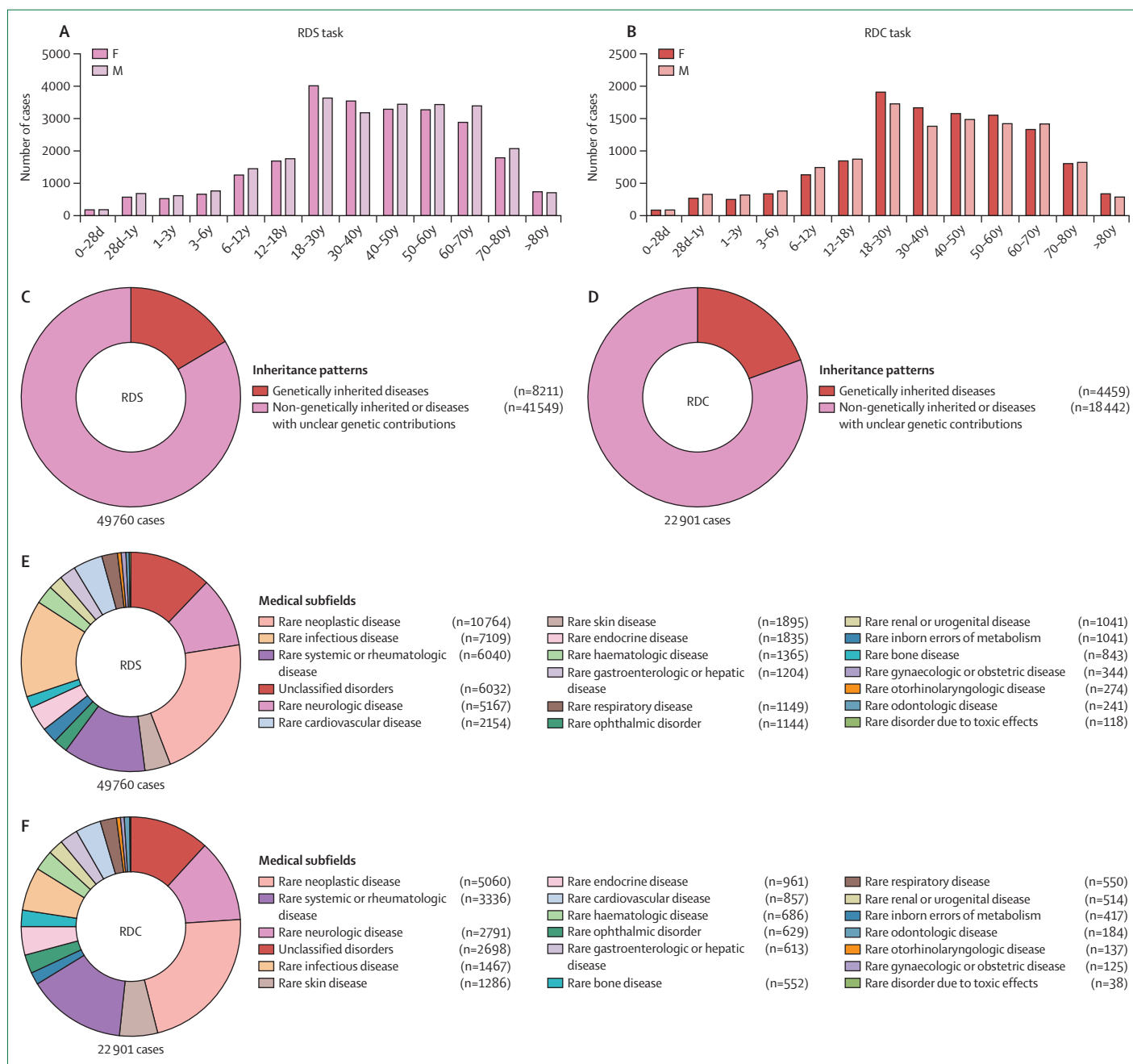


Figure 2: Overview of case distribution in RareArena

Age distribution of the cases stratified by sex in the A) RDS and B) RDC tasks. Inheritance patterns of the cases in the C) RDS and D) RDC tasks. Medical subfields of the cases in the E) RDS and F) RDC tasks. RDC=rare disease confirmation. RDS=rare disease screening.

Max: 700) in RDS and 2 (Q1: 1; Q3: 6; Max: 330) in RDC. Among the three Orphanet-defined disorder group categories (Group of Disorders, Disorder, and Subtype of disorder), the most prevalent concepts in RareArena were classified as “Disorder” (3281 [71.4%] of 4597 in RDS and 2558 [72.6%] of 3522 in RDC), covering nearly half of the disorders included in Orphanet (3281 [45.6%] of 7195 for RDS and 2558 [35.6%] of 7195 for RDC; appendix p 14).

Among Orphanet’s categories of disorders, RareArena achieved the highest coverage in the clinical syndrome category (35 [64.8%] of 54 for RDS and 31 [57.4%] of 54 for RDC).

Human evaluation

100 RareArena cases (50 each for RDS and RDC) were randomly selected for manual evaluation (table). No explicit

	Category	RDS	RDC
Leakage			
(%)	Direct leakage	0%	0%
	Indirect leakage	6%	2%
	No leakage	94%	98%
Fidelity			
Diagnosis (%)	Incorrect	0%	0%
	Partly consistent	0%	0%
	Exactly the same	100%	100%
Diagnostic test (%)	Incorrectly extracted	0%	0%
	Partly extracted	0%	4%
	Overly extracted	16%	12%
	Exactly extracted	84%	84%
Case (%)	Incorrect	0%	0%
	Partly consistent	0%	4%
	Highly consistent	8%	8%
	Exactly consistent	92%	88%
Complexity			
Pertinent negatives (%)	With negatives	86%	92%
	No negatives	14%	8%
Confounding positives (%)	With positives	40%	54%
	No positives	60%	46%
Multimorbidity (SD)		0.94 (1.32)	0.86 (1.33)
Polypharmacy (SD)		0.78 (1.17)	1.16 (1.88)
Additional hospital encounters (SD)		0.56 (0.78)	0.46 (0.70)

The evaluation is based on 100 randomly selected cases (50 each for RDS and RDC). See the subsection Human evaluation of dataset quality in the Methods for the definition of leakage, fidelity, and complexity. RDC=rare disease confirmation. RDS=rare disease screening.

Table: Human evaluation of RareArena case quality

data leakage was identified in RareArena, and only four (4%) of 100 cases contained indicative information (three [6%] of 50 for RDS and one [2%] of 50 for RDC). In terms of fidelity, all diagnoses were accurately extracted. Diagnostic tests were extracted without omissions in 98 (98%) of 100 cases, whereas in 14 (14%) of 100 cases, some non-diagnostic tests were also extracted (eight [16%] of 50 for RDS and six [12%] of 50 for RDC). Case truncation and rephrasing maintained high fidelity: eight (8%) of 100 cases missed some peripheral clinical features that did not affect the diagnosis (four [8%] of 50 for RDS and four [8%] of 50 for RDC), and only 2% had complete absence of crucial clinical features (0 of 50 for RDS and two [4%] of 50 for RDC). In terms of complexity, more than 80% of cases included descriptions of negative symptoms or clinical findings (43 [86%] of 50 for RDS and 46 [92%] of 50 for RDC), and more than 40% contained at least one confounding factor requiring careful differentiation (20 [40%] of 50 for RDS and 27 [54%] of 50 for RDC). The average number of patient morbidities was 0.94 ± 1.32 for RDS and 0.86 ± 1.33 for RDC. The average number of medications involved was 0.78 ± 1.17 for RDS and 1.16 ± 1.88 for RDC. RareArena cases also involved multiple clinical episodes, with an average of 0.56 ± 0.78 additional hospital encounters for RDC and 0.46 ± 0.70 for RDS.

Benchmarking the performance of LLMs

With this carefully curated dataset, we then evaluated the performance of current state-of-the-art LLMs in diagnosing rare diseases. Their performance on the two tasks is presented in figure 3 and the appendix (p 5; mean, standard deviation, and 95% confidence interval obtained by bootstrapping in the appendix [pp 6–7]). Generally, the RDS task was more challenging than the RDC task (adjusted $p < 0.0001$ for both top-1 recall and top-5 recall). GPT-4o consistently showed superior performance in both. For brevity, we reported only the overall recall, which is the sum of synonym matches (score 2) and hypernym matches (score 1). GPT-4o achieved a top-1 recall of 33.1% and a top-5 recall of 56.9% in RDS, whereas its performance in RDC was top-1 recall of 64.2% and top-5 recall of 85.9%. These results show the potential of GPT-4o to assist physicians in rare disease diagnosis. However, measurable room remains for improvement, underscoring the challenges posed by RareArena even for the most powerful LLMs to date. To ensure robustness, we systematically evaluated whether pretraining exposure to PubMed Central case reports introduced bias into our results. Specifically, the performance of four models (GPT-4o, two Llama3.1 variants, and Phi3-7B) were compared before and after their knowledge cutoff dates. Gemma2.9B and the Qwen2.5 variants were excluded due to the unavailability of their exact cutoff dates. Using χ^2 tests, we analysed the score distributions (pre-cutoff and post-cutoff) and found no significant differences ($p > 0.1$) across all models in each evaluation setting (RDS, RDC, top-1 recall, and top-5 recall), confirming the robustness of RareArena against potential data contamination. Detailed results, including case counts, score distributions, and p-values, are provided in the appendix (p 8).

Consistent with previous findings,³⁸ we observed an increase in model performance as the parameter size scaled up. At best, the large models (UltraMedical-70B, Llama3.1-70B, and Qwen2.5-72B) achieved a top-1 recall of 27.4% on RDS and 56.1% on RDC, and a top-5 recall of 53.2% on RDS and 81.6% on RDC. Among the small models (MedLlama-7B, UltraMedical-8B, Gemma2.9B, Phi3-7B, Llama3.1-8B, and Qwen2.5-7B), the best top-1 recall was 19.1% on RDS and 39.8% on RDC, and the best top-5 recall was 48.6% on RDS and 70.6% on RDC. Additionally, we noted that the performance differences across models of various sizes were more evident for top-1 recall than for top-5 recall. For example, in RDS, Llama3.1-70B surpassed its 8B counterpart by 83% in top-1 recall and by 15% in top-5 recall (Figure 3).

Beyond model size, domain specialisation also played a key role. Biomedical LLMs showed distinct advantages in specialised tasks, as shown by UltraMedical and MedLlama's strong performance in RDS (appendix p 5). UltraMedical-70B outperformed the best 70B general-domain models by 1.8% in top-1 recall and 5.2% in top-5 recall. Similarly, UltraMedical-8B and MedLlama-7B

outperformed the general-domain small models (Gemma2-9B, Phi3-7B, Llama3-1-8B, and Qwen2-5-7B). However, for RDC tasks, UltraMedical and MedLlama delivered performance that was only comparable to, or below, the best general-domain models.

Meanwhile, GPT-o1, a reasoning model with impressive performance in multiple medical tasks,³⁶ showed a further improvement over GPT-4o in both of the tasks. GPT-o1 achieved a top-1 recall of 38.8% on RDS and 70.9% on RDC, and a top-5 recall of 64.1% on RDS and 90.2% on RDC (appendix p 15).

Apart from model size and training domain, prompt strategies can also influence the performance of LLMs in rare disease diagnosis. Both chain-of-thought and few-shot prompting generally led to improved model performance, but the degree of enhancement varied considerably across different models and evaluation settings (appendix pp 9–10). Notably, the benefits of chain-of-thought prompting appeared more pronounced and consistent than those of few-shot prompting, likely because the fixed, randomly selected few-shot examples were often not relevant to a given case and therefore provided limited guidance for the task.

Beyond recall performance, the specificity of the models was also analysed, defined by the ratio of synonym matches (score 2) to hypernym matches (score 1), measuring the ability of the models to give specific diagnoses (appendix p 11). Overall, the models we benchmarked presented clear differences in specificity, even within the same parameter size group under the same prompt. GPT-4o showed the highest specificity, followed by Llama3-1-70B and UltraMedical-70B. Almost all models were more specific in RDC than in RDS, probably due to the highly specific information contained in diagnostic test results.

Subgroup analyses

Since each case's diagnosis was mapped to Orphanet, which provides detailed information on each disorder, each case could be characterised with its medical subfield, inheritance pattern, and disease prevalence. GPT-4o performed significantly better on genetically inherited diseases than on non-genetic diseases across both tasks (adjusted $p < 0.001$ for both top-1 recall and top-5 recall; figure 4). The performance of GPT-4o varied across different medical subfields (appendix p 16). Overall, GPT-4o achieved the highest performance in systemic or rheumatologic diseases (48.5% top-1 recall and 73.5% top-5 recall in the RDS task; 76.0% top-1 recall and 93.5% top-5 recall in the RDC task). GPT-4o's performance was lowest in neoplastic diseases for the RDS task (18.4% top-1 recall and 42.8% top-5 recall), but not for the RDC task. This observation could stem from the simple clinical symptoms of neoplastic diseases and their heavy reliance on histological results for diagnosis, which were usually excluded in the RDS task. In the RDC task, GPT-4o had the lowest top-1 recall in odontologic diseases (50.5%) and the lowest top-5 recall in otorhinolaryngologic diseases (71.5%).

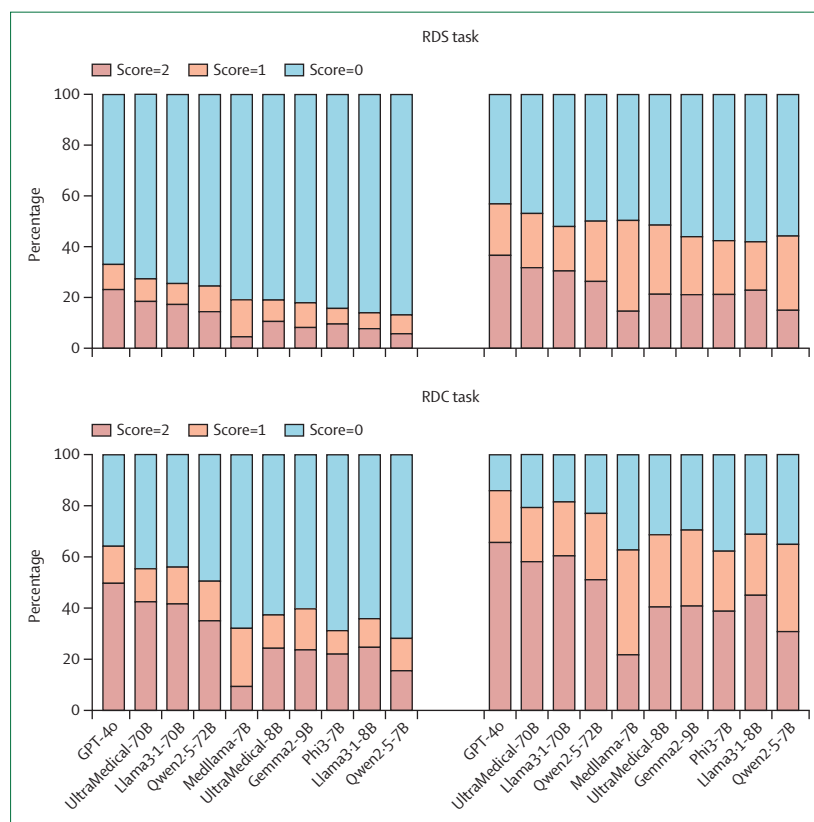


Figure 3: Performance of different large language models in RareArena

Top-1 recall and top-5 recall of different large language models in the two RareArena tasks. Score 2=synonym matches. Score 1=hypernym matches. Score 0=missing matches. RDC=rare disease confirmation. RDS=rare disease screening.

Contrary to expectations, no declining trend was observed with decreasing disease prevalence (appendix p 17). We then hypothesised that the model's capacity regarding some diseases could be positively related to the abundance of their appearance in the model's pretraining corpus. However, the pretraining corpus was either inaccessible or prohibitive to analyse, leading us to use the number of results when searching for the disease name in PubMed as a proxy in July, 2024. A clear decline in performance was observed with reduced mentions of the disease in PubMed (appendix p 17), indicating a potential correlation between the model performance and the number of encounters during training.

Discussion

In this Resource paper, we describe RareArena, the largest rare disease diagnostic benchmark to date, comprising nearly 50 000 cases across more than 4500 rare diseases, covering 3281 (45.6%) of all 7195 Orphanet disorder concepts. RareArena holds key clinical relevance, featuring task settings that reflect real-world clinical practice, authentic patient data in free text, and retained clinical complexity. With the help of this comprehensive dataset, we

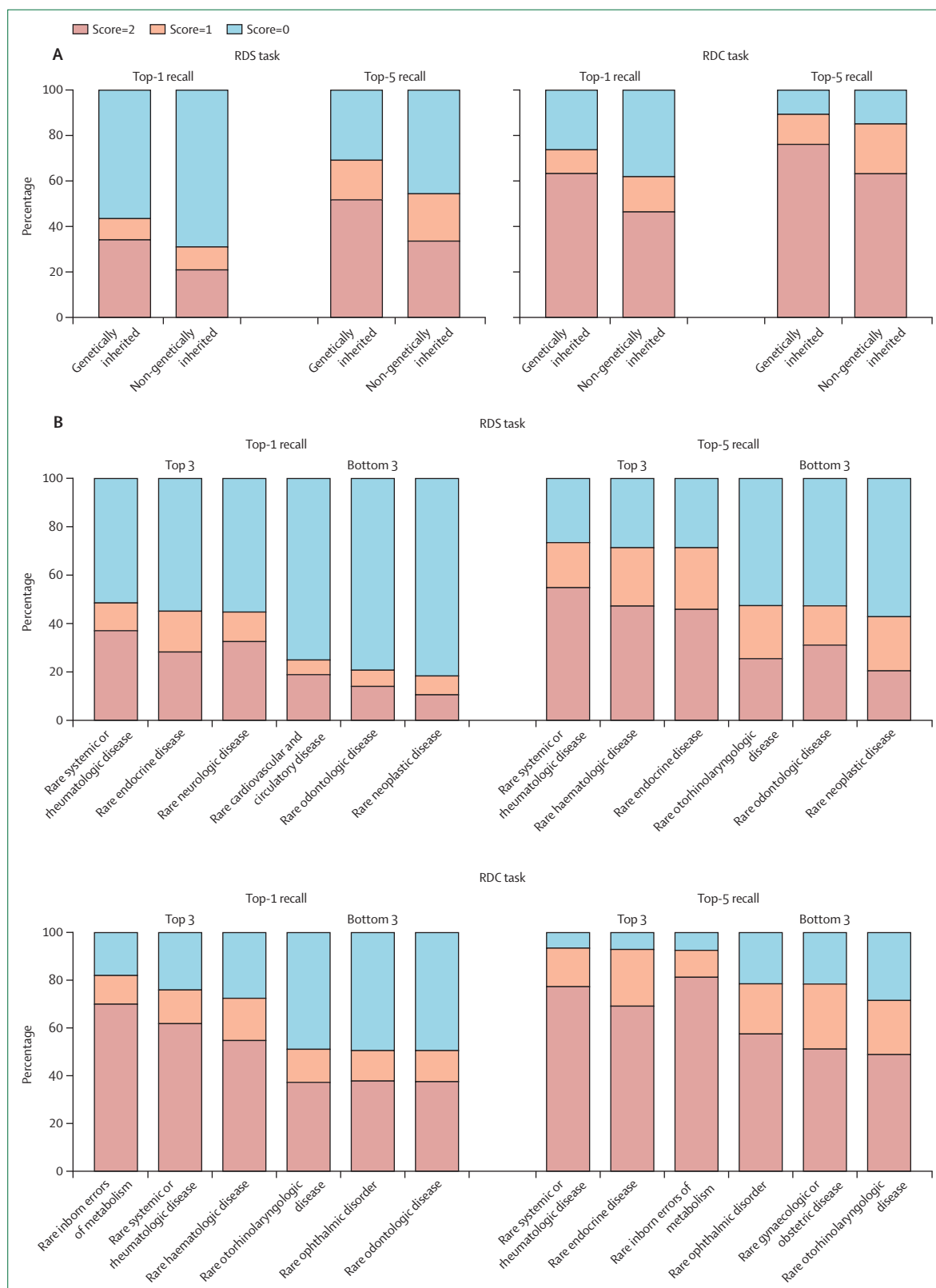


Figure 4: Subgroup analyses of the performance of GPT-4o in RareArenA

Top-1 recall and top-5 recall of GPT-4o across disease groups with A) different inheritance patterns and B) medical subfields. Panel B displays only the medical subfields with the three highest and three lowest total recall scores. Score 2=synonym matches. Score 1=hypernym matches. Score 0=missing matches. RDC=rare disease confirmation. RDS=rare disease screening.

systematically benchmarked the rare disease diagnostic capacities of leading LLMs and revealed their strong potential.

The broad coverage of diseases in RareArena enables the most comprehensive evaluation of the abilities of LLMs in rare disease diagnosis compared with previous datasets as we increased the disease coverage by several orders of magnitude, greatly enhancing the generalisability of our evaluation conclusions. Moreover, the diagnoses in RareArena were mapped to Orphanet concepts. This mapping, combined with the high disease coverage, enables more thorough subgroup analysis of LLMs' capacity in rare disease diagnosis, such as by medical subfields and rare disease prevalence. For researchers focusing on specific diseases or disorder groups, RareArena can be easily filtered into relevant subsets using Orphanet mapping for further investigations. Facilitating both overall evaluation and in-depth analysis for various purposes, RareArena holds potential value for a wide range of communities, from artificial intelligence (AI) researchers to rare disease specialists.

More importantly, we believe RareArena is notable for its strong clinical relevance and implications. First, we moved beyond examination questions to a benchmark closer to real-world clinical practice. The RDS task reflects a scenario wherein physicians need to provide possible diagnoses based on complex clinical symptoms, which usually plays a determinant role in the subsequent diagnostic success. In comparison, the RDC task represents a scenario wherein physicians need to give a confirmed diagnosis with sufficient evidence. Our benchmark results provided an initial insight into the performance of LLMs in these applications. Second, all cases in RareArena are based on real individuals, and the input for models consists of note summaries in free text, a setting closer to real-world clinical scenarios. Third, RareArena reproduces the complexity and challenges faced by physicians in the real world, incorporating appropriate negative investigations and confounding factors. Our results also indicated that RareArena could be quite challenging for even the most advanced LLMs.

The comprehensive analysis of various LLMs of different sizes provides a more detailed picture of current model capacities. The 7B parameter models, although performing well in general-domain benchmarks and sometimes comparable to their larger counterparts,³⁰ still lag behind to a great degree in RareArena. Considering the potential privacy issues for clinical AI systems and the prohibitive costs of larger models, our findings highlight the necessity for both effective and efficient models for rare diseases.³⁹ We observed that the differences between large and small models were more evident in terms of top-1 recall than top-5 recall, and that larger models generally presented higher specificity than smaller ones. These findings indicate that the models become more discriminating in diagnosing as their size scales up, which could be attributed not only to more knowledge encoded in the parameters, but also to enhanced reasoning abilities. Biomedical LLMs showed superior performance in RDS tasks, underscoring

the value of domain-specific adaptation for enhancing diagnostic capabilities in rare diseases. The superior performance and highest specificity of GPT-o1, as compared with those of other LLMs, further highlight the importance of a model's reasoning capacities in diagnostic tasks. These findings suggest a potential direction for developing more diagnostically capable models, particularly by enhancing their clinical reasoning abilities.

With current LLMs showing surprising rare disease diagnostic capacities, our analysis highlights the potential of LLMs to serve as a clinical decision support system for rare disease diagnosis. Traditional clinical decision support systems for rare diseases are usually phenotype-centred, requiring physicians to manually extract phenotypic information, which could overlook key details.^{40,41} In contrast, LLM-based clinical decision support systems can directly interpret free-text clinical notes, offering greater flexibility and contextual understanding. Integrating LLMs into clinical practice might reduce misdiagnoses of rare diseases and facilitate earlier intervention.

Nevertheless, our study presents several limitations. First, we acknowledge that our benchmark serves only as a simulation of rare disease diagnostic procedures, and the two task settings distinguished by diagnostic tests might not correspond fully to real-world settings and capture the optimal timing for using the assistance of LLMs. Because most cases contain sufficient information, our current cases could offer more details than what is typically available in real-world clinical scenarios that require decision support by an LLM, especially for RDC tasks. Second, our benchmark did not evaluate the reasoning process, whereas previous studies have shown that LLMs can generate incorrect reasoning paths despite arriving at the correct final answer.⁴² Given the potentially catastrophic consequences of flawed clinical reasoning, that future work expands evaluation to include the reasoning and explanation processes is both crucial and urgent. Third, although GPT-based evaluation of LLM responses shows high consistency with human judgements, such evaluation could still introduce inaccuracies and potential biases. Fourth, we did not test more advanced prompt strategies, such as dynamic few-shot prompting and retrieval augmented generation, which could further enhance LLM performance. As our work focused primarily on dataset curation and benchmarking base model capabilities, and implementing these advanced approaches would introduce numerous confounding variables (eg, few-shot example selection, retriever performance, corpus design), we leave this investigation to future research. Fifth, although many rare diseases require multimodal diagnostic tests (eg, imaging, histopathology, and genetic analyses), we only used free-text inputs. Although these texts typically incorporate key findings from such tests, this unimodal approach might miss detailed information and pose some limitations on clinical realism, particularly for the RDC task. Developing multimodal benchmarks and models for rare disease diagnosis remains a promising direction for

Search strategy and selection criteria

We searched PubMed, Google Scholar, and MEDLINE for peer-reviewed journal articles, conference papers, and preprint articles published from database inception to Aug 5, 2024, without language restrictions, using the terms “rare disease” AND “diagnosis” AND “large language model” OR “LLM” OR “benchmark” OR “dataset”. We also checked the reference lists of related publications. We did not identify papers on benchmarking the performance of large language models (LLMs) across a comprehensive spectrum of rare diseases.

future work. Sixth, we acknowledge the absence of sex or gender subgroup analyses. The sex or gender information in our dataset was extracted directly from the original case reports without standardised definitions or assessment methods; therefore, we did not perform subgroup analyses by sex or gender. Race and ethnicity were not considered in this study as these variables were largely absent from the original case reports.

To our knowledge, our study presents the most comprehensive rare disease diagnostic dataset to date, with high clinical relevance in real-world settings. Using this dataset, we offered an in-depth assessment of the capabilities of various LLMs in diagnosing rare diseases and thoroughly examined the differences in their performance across different diseases. We found that GPT-4o consistently surpassed other models, particularly excelling in genetically inherited and systemic or rheumatologic conditions. Our study underscores the potential of LLMs to assist in accurate rare disease diagnoses in clinical practice in different settings globally, while highlighting the importance of further improvements, especially for developing more efficient models and enhanced clinical reasoning capabilities.

Contributors

HC, ZZ, and SZhou conceived and planned this study. HC and ZZ analysed and interpreted the data and drafted the manuscript. SH, HC, and XJ evaluated the case quality. All authors contributed to the intellectual development of this study and provided critical feedback on the manuscript. All authors had full access to and verified all data in the study (including the data presented in the manuscript and uploaded to GitHub and HuggingFace) and had final responsibility for the decision to submit for publication. TYW took responsibility for the integrity of the data and the accuracy of the data analysis.

Declaration of interests

TYW received consulting fees from Aldropika Therapeutics, Bayer, Boehringer Ingelheim, Genentech, Iveric Bio, Novartis, Plano, Oxurion, Roche, Sanofi, and Shanghai Henlius Biotech; and is an inventor, patent holder, and cofounder of the start-up companies EyRIS and Visre. All other authors declare no competing interests.

Data sharing

The dataset has been released under CC BY-NC-SA licence on both GitHub and HuggingFace.

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For the RareArena dataset on GitHub see <https://github.com/zhao-zy15/RareArena>

For the RareArena dataset on HuggingFace see <https://huggingface.co/datasets/THUMedInfo/RareArena>

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